Section 2.1

Question 2

(a)

Let S(t) represent the number of steps recorded t minutes after 3:00 PM. The slope of a secant line over an interval [a,b] is

$$\frac{\Delta S}{\Delta t} = \frac{S(b) - S(a)}{b - a}$$

This slope represents the average walking rate in steps per minute.

(i) Interval [0,40]

$$\frac{S(40) - S(0)}{40 - 0} = \frac{7398 - 3438}{40} = \frac{3960}{40} = \boxed{99 \text{ steps/min}}$$

(ii) Interval [10,20]

$$\frac{S(20) - S(10)}{20 - 10} = \frac{5622 - 4559}{10} = \frac{1063}{10} = \boxed{106.3 \text{ steps/min}}$$

(iii) Interval [20,30]

$$\frac{S(30) - S(20)}{30 - 20} = \frac{6536 - 5622}{10} = \frac{914}{10} = \boxed{91.4 \text{ steps/min}}$$

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b

The number of steps is given as a function of time t, measured in minutes after 3:00 PM.

Since 3:20 PM is **20 minutes after 3:00 PM**, we estimate the student's walking pace at $t = 20$ by averaging the slopes of two secant lines on either side of $t = 20$.

Secant slopes from part (a)

- On the interval $[10, 20]$:

$$\frac{5622 - 4559}{20 - 10} = \frac{1063}{10} = 106.3 \text{ steps/min}$$

- On the interval $[20, 30]$:

$$\frac{6536-5622}{30-20}=\frac{914}{10}=91.4\ \text{steps/min}$$

Average of the two secant slopes

$$\frac{106.3+91.4}{2}=98.85\ \text{steps/min}$$

Question 5

Given the height function (in feet):

$$y(t)=275-16t^2$$

(a)

Average velocity over the interval ($[4,\ 4+\Delta t]$) is

$$v_{\text{avg}}=\frac{y(4+\Delta t)-y(4)}{\Delta t}$$

First compute:

$$y(4)=275-16(4)^2=19$$

i. ($\Delta t=0.1$)

$$y(4.1)=275-16(4.1)^2=6.04$$

$$v_{\text{avg}}=\frac{6.04-19}{0.1}=\boxed{-129.6\ \text{ft/s}}$$

ii. ($\Delta t=0.05$)

$$y(4.05)=275-16(4.05)^2=12.56$$

$$v_{\text{avg}}=\frac{12.56-19}{0.05}=\boxed{-128.8\ \text{ft/s}}$$

iii. ($\Delta t=0.01$)

$$y(4.01)=275-16(4.01)^2=17.7184$$

$$v_{\text{avg}}=\frac{17.7184-19}{0.01}=\boxed{-128.16\ \text{ft/s}}$$

(b)

As ($\Delta t\to 0$), the average velocities approach

$$\boxed{-128\ \text{ft/s}}$$

(The negative sign indicates downward motion.)

Question 7

(a.) Average velocity on an interval $[a, b]$ is

$$v_{\text{avg}} = \frac{s(b) - s(a)}{b - a}.$$

From the table (position in feet, time in seconds):

$$s(2) = 20.6 \text{ ft}, \quad s(3) = 46.5 \text{ ft}, \quad s(4) = 79.2 \text{ ft}, \quad s(5) = 124.8 \text{ ft}, \\ s(6) = 176.7 \text{ ft}.$$

(i.) Interval $[2, 4]$:

$$\begin{aligned} v_{\text{avg}}[2,4] &= \frac{s(4) - s(2)}{4 - 2} \\ &= \frac{79.2 \text{ ft} - 20.6 \text{ ft}}{4 \text{ s} - 2 \text{ s}} \\ &= \frac{58.6 \text{ ft}}{2 \text{ s}} \\ &= 29.3 \text{ ft/s}. \end{aligned}$$

$$\boxed{v_{\text{avg}}[2,4] = 29.3 \text{ ft/s}}$$

(ii.) Interval $[3, 4]$:

$$\begin{aligned} v_{\text{avg}}[3,4] &= \frac{s(4) - s(3)}{4 - 3} \\ &= \frac{79.2 \text{ ft} - 46.5 \text{ ft}}{4 \text{ s} - 3 \text{ s}} \\ &= \frac{32.7 \text{ ft}}{1 \text{ s}} \\ &= 32.7 \text{ ft/s}. \end{aligned}$$

$$\boxed{v_{\text{avg}}[3,4] = 32.7 \text{ ft/s}}$$

(iii.) Interval $[4, 5]$:

$$\begin{aligned} v_{\text{avg}}[4,5] &= \frac{s(5) - s(4)}{5 - 4} \\ &= \frac{124.8 \text{ ft} - 79.2 \text{ ft}}{5 \text{ s} - 4 \text{ s}} \\ &= \frac{45.6 \text{ ft}}{1 \text{ s}} \\ &= 45.6 \text{ ft/s}. \end{aligned}$$

$$\boxed{v_{\text{avg}}[4,5] = 45.6\ \text{ft/s}}$$

(iv.) Interval $[4, 6]$:

$$\begin{aligned} v_{\text{avg}}[4,6] &= \frac{s(6) - s(4)}{6 - 4} \\ &= \frac{176.7\ \text{ft} - 79.2\ \text{ft}}{6\ \text{s} - 4\ \text{s}} \\ &= \frac{97.5\ \text{ft}}{2\ \text{s}} \\ &= 48.75\ \text{ft/s}. \end{aligned}$$

$$\boxed{v_{\text{avg}}[4,6] = 48.75\ \text{ft/s}}$$

(b.) Use the graph of s as a function of t to estimate the instantaneous velocity when $t = 3$.

Use average velocities on intervals closest to $t = 3$:

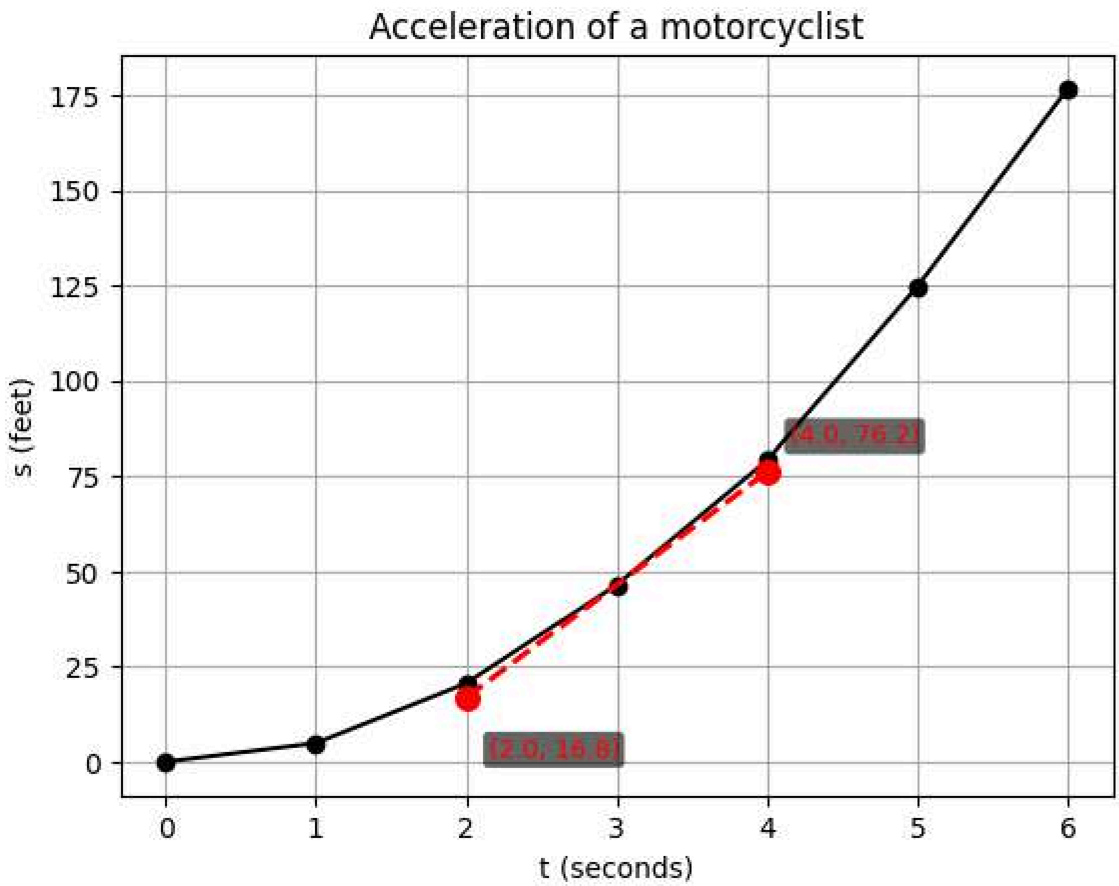
$$\begin{aligned} v_{\text{avg}}[2,3] &= \frac{s(3) - s(2)}{3 - 2} \\ &= \frac{46.5\ \text{ft} - 20.6\ \text{ft}}{3\ \text{s} - 2\ \text{s}} \\ &= \frac{25.9\ \text{ft}}{1\ \text{s}} \\ &= 25.9\ \text{ft/s}, \end{aligned}$$

$$\begin{aligned} v_{\text{avg}}[3,4] &= \frac{s(4) - s(3)}{4 - 3} \\ &= \frac{79.2\ \text{ft} - 46.5\ \text{ft}}{4\ \text{s} - 3\ \text{s}} \\ &= \frac{32.7\ \text{ft}}{1\ \text{s}} \\ &= 32.7\ \text{ft/s}. \end{aligned}$$

A reasonable estimate for the instantaneous velocity at $t = 3$ is the midpoint of these nearby slopes:

$$v(3) \approx \frac{25.9 + 32.7}{2}\ \text{ft/s} = 29.3\ \text{ft/s}.$$

$$\boxed{v(3) \approx 29.3\ \text{ft/s}}$$



(b.) Use the graph of s as a function of t to estimate the instantaneous velocity when $t = 3$.

To estimate the instantaneous velocity at $t = 3$, we consider the slope of the tangent line to the graph of $s(t)$ at $t = 3$. Using the graph of the data points

$$t = [0, 1, 2, 3, 4, 5, 6], \quad s = [0, 4.9, 20.6, 46.5, 79.2, 124.8, 176.7],$$

a tangent line drawn at the point $(3, 46.5)$ has slope approximately 29.7.

Thus, the instantaneous velocity at $t = 3$ is estimated to be

$$\boxed{v(3) \approx 29.7 \text{ ft/s}}.$$

Section 2.2

Question 6

(a.) Left-hand limit

$$\boxed{\lim_{x \to -3^-} h(x) = 4}$$

Exists: Yes.

(b.) Right-hand limit

$$\boxed{\lim_{x \to -3^+} h(x) = 4}$$

Exists: Yes.

(c.) Two-sided limit

$$\boxed{\lim_{x \to -3} h(x) = 4}$$

Exists: Yes.

(d.) Function value

$$\boxed{h(-3)\ \text{DNE}}$$

Exists: No.

Reason: There is no filled point at $x = -3$.

(e.) Left-hand limit

$$\boxed{\lim_{x \to 0^-} h(x) = 1}$$

Exists: Yes.

(f.) Right-hand limit

$$\boxed{\lim_{x \to 0^+} h(x) = -1}$$

Exists: Yes.

(g.) Two-sided limit

$$\boxed{\lim_{x \to 0} h(x)\ \text{DNE}}$$

Exists: No.

Reason: The left-hand limit and right-hand limit are not equal.

(h.) Function value

$$\boxed{h(0) = 1}$$

Exists: Yes.

(i.) Two-sided limit

$$\boxed{\lim_{x \to 2} h(x) = 2}$$

Exists: Yes.

(j.) Function value

$$\boxed{h(2)\ \text{DNE}}$$

Exists: No.

Reason: There is an open circle at $x = 2$ and no filled point.

(k.) Right-hand limit

$$\boxed{\lim_{x \to 5^+} h(x) = 3}$$

Exists: Yes.

(l.) Left-hand limit

$$\boxed{\lim_{x \to 5^-} h(x)\ \text{DNE}}$$

Exists: No.

Reason: As x approaches 5 from the left, $h(x)$ oscillates between approximately 2 and 4 and does not approach a single value.

Question 10

A patient receives a 150-mg injection of a drug every 4 hours. The graph shows the amount $f(t)$ of the drug in the bloodstream after t hours.

Left-hand limit

$$\lim_{t \to 12^-} f(t)$$

$$\boxed{150}$$

Right-hand limit

$$\lim_{t \to 12^+} f(t)$$

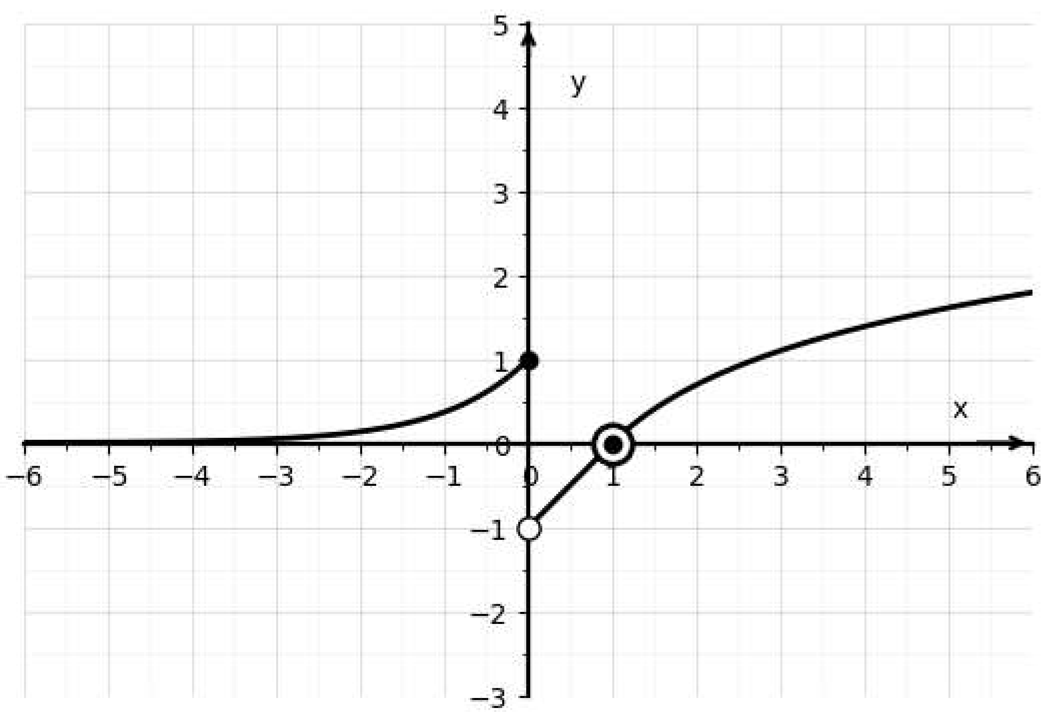
$$\boxed{300}$$

Interpretation

- $\lim_{t \to 12^-} f(t)$ is the amount of drug just before the injection at $t = 12$.
- $\lim_{t \to 12^+} f(t)$ is the amount of drug immediately after the injection at $t = 12$.

Question 11

$$f(x) = \begin{cases} e^x, & x \le 0, \\ x - 1, & 0 < x < 1, \\ \ln(x), & x \ge 1 \end{cases}$$



Given the piecewise function we determine the values of a for which $\lim_{x \to a} f(x)$ exists.

Each expression defining f is continuous on its interval. Therefore, $\lim_{x \to a} f(x)$ exists for all a in the interior of each interval. The limit can fail to exist only at the transition points $x = 0$ and $x = 1$.

At $x = 0$:

$$\lim_{x \to 0^-} f(x) = e^0 = 1, \qquad \lim_{x \to 0^+} f(x) = 0 - 1 = -1$$

Since the one-sided limits are not equal, $\lim_{x \to 0} f(x)$ does not exist.

At $x = 1$:

$$\lim_{x\to 1^-} f(x) = 1 - 1 = 0, \qquad \lim_{x\to 1^+} f(x) = \ln(1) = 0$$

Since the one-sided limits are equal, $\lim_{x\to 1} f(x)$ exists.

$$\boxed{\lim_{x\to a} f(x) \text{ exists for all } a \neq 0}$$

Question 19

Numerical evaluation near $x = 3$

Let

$$f(x) = \frac{x^2 - 3x}{x^2 - 9}, \quad x \neq 3.$$

Evaluate the function at values close to 3, rounding to six decimal places.

$$f(3.1) = \frac{(3.1)^2 - 3(3.1)}{(3.1)^2 - 9} = \frac{9.61 - 9.3}{9.61 - 9} = \boxed{0.508197}$$

$$f(3.05) = \frac{(3.05)^2 - 3(3.05)}{(3.05)^2 - 9} = \frac{9.3025 - 9.15}{9.3025 - 9} = \boxed{0.504132}$$

$$f(3.01) = \frac{(3.01)^2 - 3(3.01)}{(3.01)^2 - 9} = \frac{9.0601 - 9.03}{9.0601 - 9} = \boxed{0.500832}$$

$$f(3.001) = \frac{(3.001)^2 - 3(3.001)}{(3.001)^2 - 9} = \frac{9.006001 - 9.003}{9.006001 - 9} = \boxed{0.500083}$$

$$f(3.0001) = \frac{(3.0001)^2 - 3(3.0001)}{(3.0001)^2 - 9} = \frac{9.00060001 - 9.0003}{9.00060001 - 9} = \boxed{0.500008}$$

$$f(2.9) = \frac{(2.9)^2 - 3(2.9)}{(2.9)^2 - 9} = \frac{8.41 - 8.7}{8.41 - 9} = \boxed{0.491525}$$

$$f(2.95) = \frac{(2.95)^2 - 3(2.95)}{(2.95)^2 - 9} = \frac{8.7025 - 8.85}{8.7025 - 9} = \boxed{0.495798}$$

$$f(2.99) = \frac{(2.99)^2 - 3(2.99)}{(2.99)^2 - 9} = \frac{8.9401 - 8.97}{8.9401 - 9} = \boxed{0.499165}$$

$$f(2.999) = \frac{(2.999)^2 - 3(2.999)}{(2.999)^2 - 9} = \frac{8.994001 - 8.997}{8.994001 - 9} = \boxed{0.499917}$$

From both the left and the right, the function values approach 0.5.

$$\boxed{\frac{1}{2}}$$

Question 32

$$\lim_{x \to 3^-} \frac{\sqrt{x}}{(x-3)^5}$$

- As $x \to 3^-$, $\sqrt{x} \to \sqrt{3}$ (finite and positive)
- $(x-3)^5 \to 0^-$ (odd power preserves sign)

Thus the quotient decreases without bound.

$$\boxed{-\infty}$$

Question 38

$$\lim_{x \to 3^-} \frac{x^2+4x}{x^2-2x-3}$$

Factor:

$$\frac{x(x+4)}{(x-3)(x+1)}$$

Sign analysis as $x \to 3^-$:

- Numerator > 0
- Denominator < 0

Therefore the expression decreases without bound.

$$\boxed{-\infty}$$

Question 40

$$\lim_{x \to 0^+} \left(\frac{1}{x} - \ln x\right)$$

As $x \to 0^+$:

$$\frac{1}{x} \to +\infty, \qquad \ln x \to -\infty$$

Thus

$$\frac{1}{x} - \ln x \to +\infty$$

Section 2.3

Question 6

Evaluate the limit:

$$\lim_{t \to 7} \frac{3t^2+1}{t^2-5t+2}$$

By the **Quotient Law**, provided the denominator limit is nonzero,

$$\lim_{t \to 7} \frac{3t^2+1}{t^2-5t+2} = \frac{\lim_{t\to 7}(3t^2+1)}{\lim_{t\to 7}(t^2-5t+2)}$$

Numerator

Using the **Sum Law** and **Constant Multiple Law**,

$$\lim_{t\to 7}(3t^2+1) = \lim_{t\to 7}(3t^2) + \lim_{t\to 7}(1) = 3\lim_{t\to 7}(t^2) + 1$$

Using the **Power of x Law** and **Constant Law**,

$$= 3(7^2) + 1 = 148$$

Denominator

Using the **Difference Law**, **Sum Law**, and **Constant Multiple Law**,

$$\lim_{t\to 7}(t^2-5t+2) = \lim_{t\to 7}(t^2) - \lim_{t\to 7}(5t) + \lim_{t\to 7}(2)$$

Using the **Power of x Law**, **Identity Law**, and **Constant Law**,

$$= 7^2 - 5(7) + 2 = 16$$

Validity of the Quotient Law

Since

$$\lim_{t\to 7}(t^2-5t+2) = 16 \neq 0,$$

the Quotient Law applies

$$\boxed{\frac{37}{4}}$$

Question 10

(a.) What is wrong with the equation

$$\frac{x^2 + x - 6}{x - 2} = x + 3\,?$$

Factor the numerator:

$$x^2 + x - 6 = (x + 3)(x - 2)$$

So for $x \neq 2$,

$$\frac{x^2 + x - 6}{x - 2} = \frac{(x + 3)(x - 2)}{x - 2} = x + 3$$

But at $x = 2$ the left-hand side is **undefined** (division by 0), while $x + 3$ **is** defined. So the equation is not true "for all x"; it is only true for $x \neq 2$.

$$\boxed{\text{The left side is undefined at } x = 2, \text{ so it cannot equal } x + 3 \text{ there.}}$$

(b.) Explain why

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2} = \lim_{x \to 2}(x + 3)$$

is correct.

Because the two functions are equal for all x sufficiently close to 2 (except at $x = 2$ itself):

$$\frac{x^2 + x - 6}{x - 2} = x + 3 \quad \text{for } x \neq 2$$

$$\boxed{\text{Limits only depend on values near the point, not at the point, so the limits agree.}}$$

$$\lim_{x \to 2} \frac{x^2 + x - 6}{x - 2} = \lim_{x \to 2}(x + 3) = 5$$

Question 16

Evaluate (if it exists):

$$\lim_{x \to 4} \frac{x^2 + 3x}{x^2 - x - 12}$$

Factor

$$\frac{x^2+3x}{x^2-x-12}=\frac{x(x+3)}{(x-4)(x+3)}$$

Cancel the common factor $(x+3)$ (valid when $x \neq -3$)

$$\frac{x(x+3)}{(x-4)(x+3)}=\frac{x}{x-4} \quad (x \neq -3)$$

Now check one-sided behavior as $x \to 4$

(a.) Left-hand limit

$$\lim_{x\to 4^-}\frac{x^2+3x}{x^2-x-12}=\lim_{x\to 4^-}\frac{x}{x-4}$$

As $x \to 4^-$, we have $x \approx 4 > 0$ and $x-4<0$, so $\frac{x}{x-4} \to -\infty$

$$\lim_{x\to 4^-}\frac{x^2+3x}{x^2-x-12}=-\infty$$

(b.) Right-hand limit

$$\lim_{x\to 4^+}\frac{x^2+3x}{x^2-x-12}=\lim_{x\to 4^+}\frac{x}{x-4}$$

As $x \to 4^+$, we have $x \approx 4 > 0$ and $x-4>0$, so $\frac{x}{x-4} \to +\infty$

$$\lim_{x\to 4^+}\frac{x^2+3x}{x^2-x-12}=+\infty$$

Since the one-sided limits are not equal, the two-sided limit does not exist

$$\boxed{\lim_{x\to 4}\frac{x^2+3x}{x^2-x-12}\ \mathrm{DNE}}$$

Question 18

Evaluate (if it exists):

$$\lim_{x\to -5}\frac{2x^2+9x-5}{x^2-25}$$

Factor

$$2x^2+9x-5=(2x-1)(x+5), \qquad x^2-25=(x-5)(x+5)$$

Cancel $(x+5)$ (valid when $x \neq -5$)

$$\frac{2x^2+9x-5}{x^2-25} = \frac{(2x-1)(x+5)}{(x-5)(x+5)} = \frac{2x-1}{x-5}$$

Now substitute $x = -5$

$$\lim_{x\to -5} \frac{2x-1}{x-5} = \frac{2(-5)-1}{-5-5} = \frac{-11}{-10} = \frac{11}{10}$$

Therefore the limit exists

$$\boxed{\frac{11}{10}}$$

Question 19

Evaluate (if it exists):

$$\lim_{t\to 3} \frac{t^3-27}{t^2-9}$$

Factor

$$t^3-27 = (t-3)(t^2+3t+9), \qquad t^2-9 = (t-3)(t+3)$$

Cancel $(t-3)$ (valid when $t \neq 3$)

$$\frac{t^3-27}{t^2-9} = \frac{(t-3)(t^2+3t+9)}{(t-3)(t+3)} = \frac{t^2+3t+9}{t+3}$$

Now substitute $t = 3$

$$\lim_{t\to 3} \frac{t^2+3t+9}{t+3} = \frac{9+9+9}{6} = \frac{27}{6} = \frac{9}{2}$$

Therefore the limit exists

$$\boxed{\frac{9}{2}}$$

Question 26

Evaluate (if it exists):

$$\lim_{h\to 0} \frac{\dfrac{1}{-2+h} - \dfrac{1}{-2}}{h}$$

Rewrite the numerator

$$\frac{1}{-2+h} - \frac{1}{-2} = \frac{1}{-2+h} + \frac{1}{2}$$

Combine the fractions

$$\frac{1}{-2+h} + \frac{1}{2} = \frac{2+(-2+h)}{2(-2+h)} = \frac{h}{2(-2+h)}$$

Divide by h

$$\frac{\frac{h}{2(-2+h)}}{h} = \frac{1}{2(-2+h)}$$

Take the limit by direct substitution

$$\lim_{h\to 0} \frac{1}{2(-2+h)} = \frac{1}{2(-2)} = -\frac{1}{4}$$

Therefore the limit exists

$$\boxed{-\frac{1}{4}}$$

Question 30

Evaluate (if it exists):

$$\lim_{x\to 2} \frac{x^2-4x+4}{x^4-3x^2-4}$$

Factor the numerator

$$x^2-4x+4 = (x-2)^2$$

Factor the denominator

$$x^4-3x^2-4 = (x^2-4)(x^2+1) = (x-2)(x+2)(x^2+1)$$

Cancel one factor of $(x-2)$ (valid when $x \neq 2$)

$$\frac{(x-2)^2}{(x-2)(x+2)(x^2+1)} = \frac{x-2}{(x+2)(x^2+1)}$$

Take the limit by direct substitution

$$\lim_{x\to 2} \frac{x-2}{(x+2)(x^2+1)} = \frac{2-2}{(2+2)(2^2+1)} = \frac{0}{(4)(5)} = 0$$

Therefore the limit exists

$$\boxed{0}$$

Question 38

Evaluate:

$$\lim_{x \to 0} \sqrt{x^3 + x^2 \sin\left(\frac{\pi}{x}\right)}$$

For all $x \neq 0$, we know

$$-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1 \quad \Rightarrow \quad x^2 \sin\left(\frac{\pi}{x}\right) \leq x^2 \ \ (\text{since } x^2 \geq 0)$$

So

$$x^3 + x^2 \sin\left(\frac{\pi}{x}\right) \leq x^3 + x^2 = x^2(x+1)$$

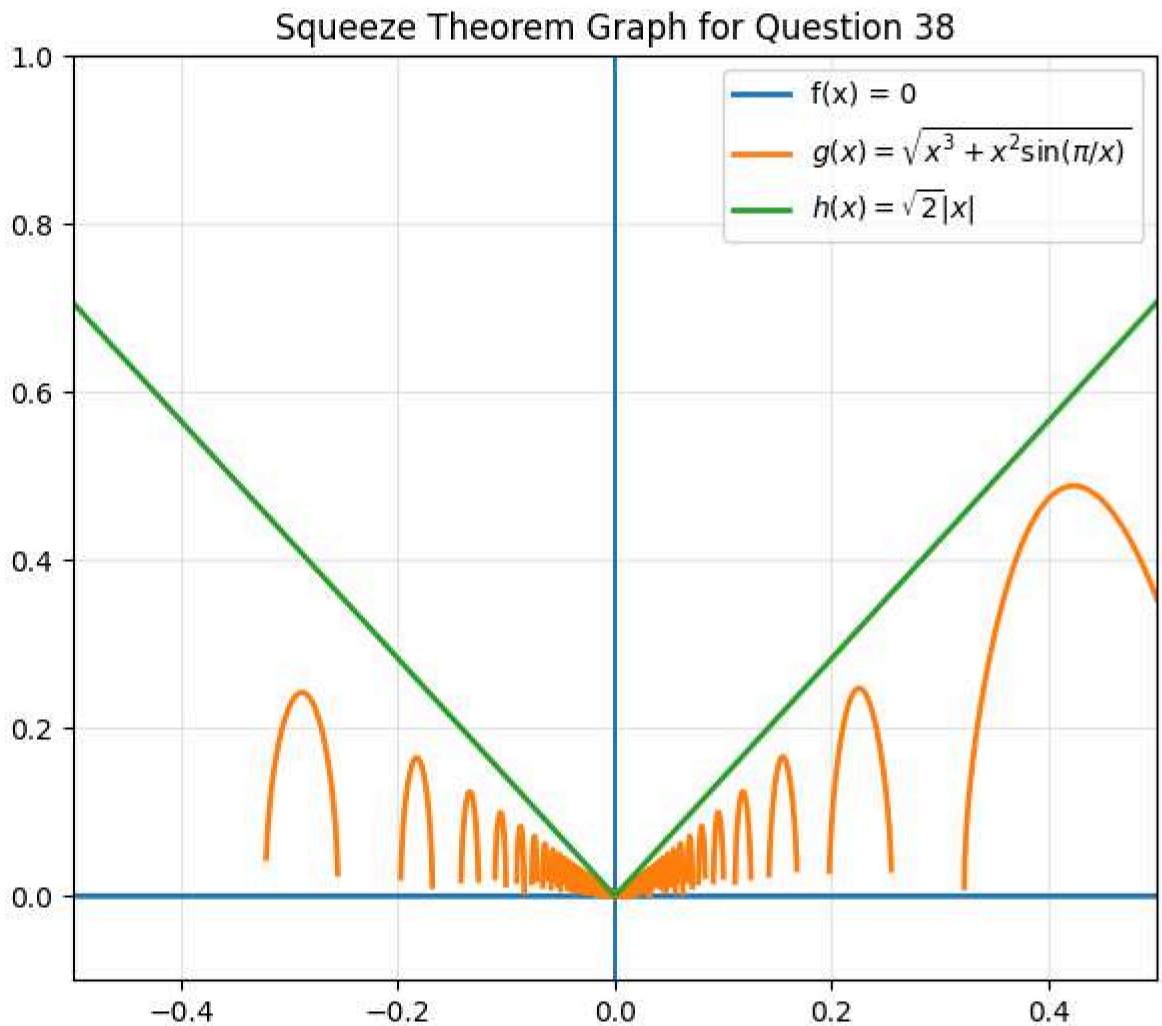
Whenever the square-root expression is defined (i.e., the inside is ≥ 0), we have:

$$0 \leq \sqrt{x^3 + x^2 \sin\left(\frac{\pi}{x}\right)} \leq \sqrt{x^2(x+1)} = |x|\sqrt{x+1}$$

As $x \to 0$, $|x|\sqrt{x+1} \to 0$. By the Squeeze Theorem,

$$\lim_{x \to 0} \sqrt{x^3 + x^2 \sin\left(\frac{\pi}{x}\right)} = 0$$

$$\boxed{0}$$



Section 2.4

Question 2

We want a number $\delta > 0$ such that if $0 < |x-3| < \delta$, then $|f(x) - 2| < 0.5$

The condition $|f(x) - 2| < 0.5$ means

$$1.5 < f(x) < 2.5$$

From the graph, the horizontal lines $y = 1.5$ and $y = 2.5$ intersect the curve at approximately

$$x = 2.6 \quad \text{and} \quad x = 3.8$$

So we need $(3-\delta,\ 3+\delta) \subset (2.6,\ 3.8)$

The largest allowable δ is the smaller distance to the endpoints

$$\delta = \min\{3 - 2.6,\ 3.8 - 3\} = \min\{0.4,\ 0.8\} = 0.4$$

$$\boxed{\delta = 0.4}$$

Question 14

Given that

$$\lim_{x \to 2}(5x - 7) = 3$$

find values of δ corresponding to $\varepsilon = 0.1,\ 0.05,\ 0.01$

Start from the ε-condition

$$|(5x - 7) - 3| < \varepsilon$$

Simplify

$$|(5x - 7) - 3| = |5x - 10| = 5|x - 2|$$

So it is enough to require

$$5|x - 2| < \varepsilon \quad \Longrightarrow \quad |x - 2| < \frac{\varepsilon}{5}$$

$$\boxed{\delta = \frac{\varepsilon}{5}}$$

Now plug in the requested ε values

- If $\varepsilon = 0.1$, then $\delta = \dfrac{0.1}{5} = 0.02$, so

$$\boxed{\delta = 0.02}$$

- If $\varepsilon = 0.05$, then $\delta = \dfrac{0.05}{5} = 0.01$, so

$$\boxed{\delta = 0.01}$$

- If $\varepsilon = 0.01$, then $\delta = \dfrac{0.01}{5} = 0.002$, so

$$\boxed{\delta = 0.002}$$

Question 16

Prove using the ε-δ definition that

$$\lim_{x \to 2}(2 - 3x) = -4$$

Let $\varepsilon > 0$ be given

We want $\delta > 0$ such that if $0 < |x-2| < \delta$, then

$$|(2-3x)-(-4)| < \varepsilon$$

Simplify the left-hand side

$$|(2-3x)+4| = |6-3x| = |-3(x-2)| = 3|x-2|$$

So it is enough to require

$$3|x-2| < \varepsilon \quad \Longrightarrow \quad |x-2| < \frac{\varepsilon}{3}$$

$$\boxed{\delta = \frac{\varepsilon}{3}}$$

Then $0 < |x-2| < \delta$ implies $|(2-3x)-(-4)| = 3|x-2| < 3\delta = \varepsilon$, as required

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Question 17

Prove using the ε-δ definition that

$$\lim_{x \to -2} (-2x+1) = 5$$

Let $\varepsilon > 0$ be given

We want $\delta > 0$ such that if $0 < |x-(-2)| = |x+2| < \delta$, then

$$|(-2x+1)-5| < \varepsilon$$

Simplify

$$|(-2x+1)-5| = |-2x-4| = |-2(x+2)| = 2|x+2|$$

So it is enough to require

$$2|x+2| < \varepsilon \quad \Longrightarrow \quad |x+2| < \frac{\varepsilon}{2}$$

$$\boxed{\delta = \frac{\varepsilon}{2}}$$

Then $0 < |x+2| < \delta$ implies $|(-2x+1)-5| = 2|x+2| < 2\delta = \varepsilon$, as required

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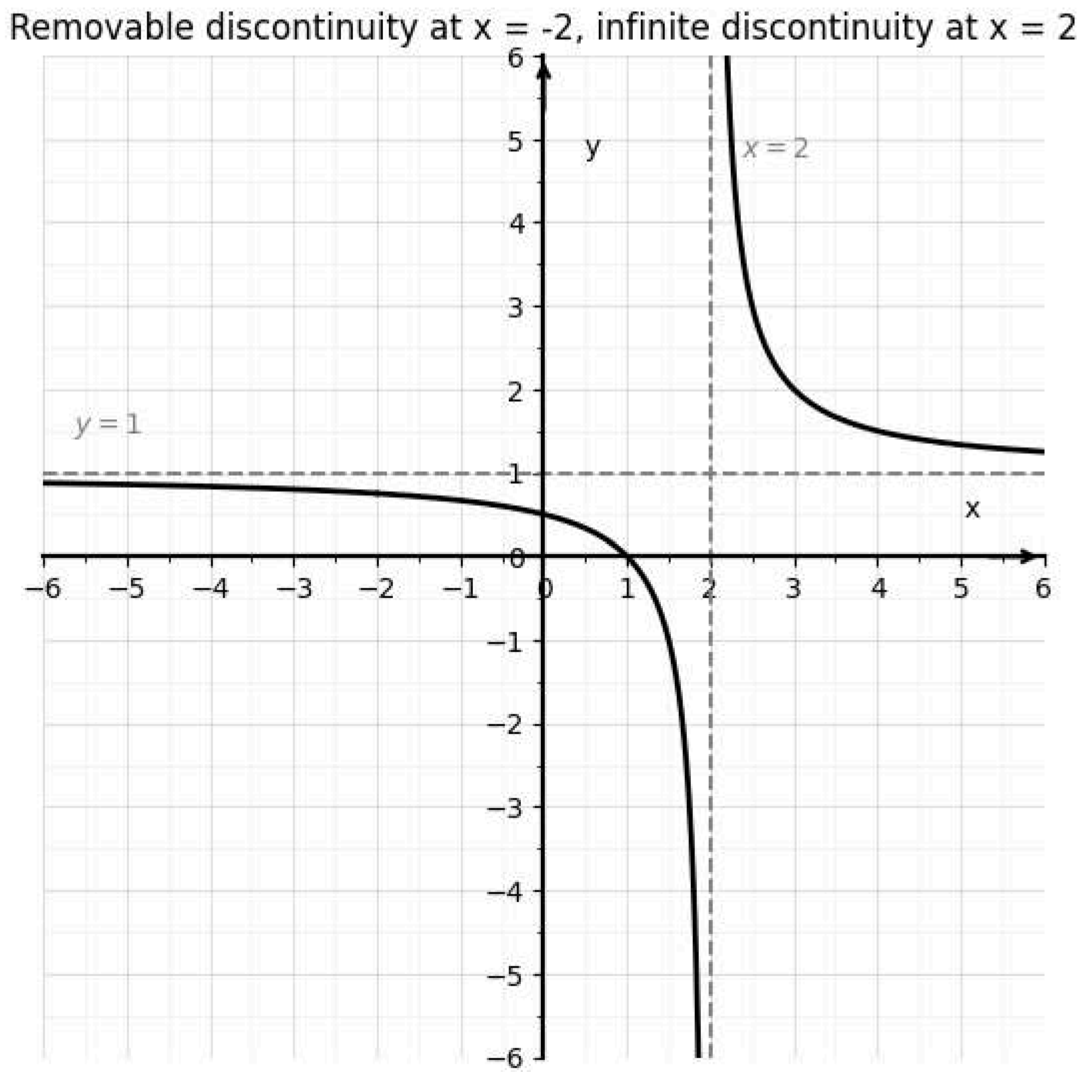
Section 2.5

Question 7

Sketch the graph of a function f that is defined on $\mathbb{R}$ and continuous except for:

- a **removable discontinuity** at $x = -2$
- an **infinite discontinuity** at $x = 2$



Question 20

The function is defined by

$$f(x) = \begin{cases} \dfrac{1}{x+2}, & x \neq -2, \\ 1, & x = -2 \end{cases}$$

with $a = -2$

Evaluate continuity at $x = -2$

Two-sided limit

$$\lim_{x \to -2} f(x) = \lim_{x \to -2} \frac{1}{x+2}$$

As $x \to -2^-$, we have $x + 2 < 0$, so

$$\lim_{x \to -2^-} \frac{1}{x+2} = -\infty$$

As $x \to -2^+$, we have $x + 2 > 0$, so

$$\lim_{x \to -2^+} \frac{1}{x+2} = +\infty$$

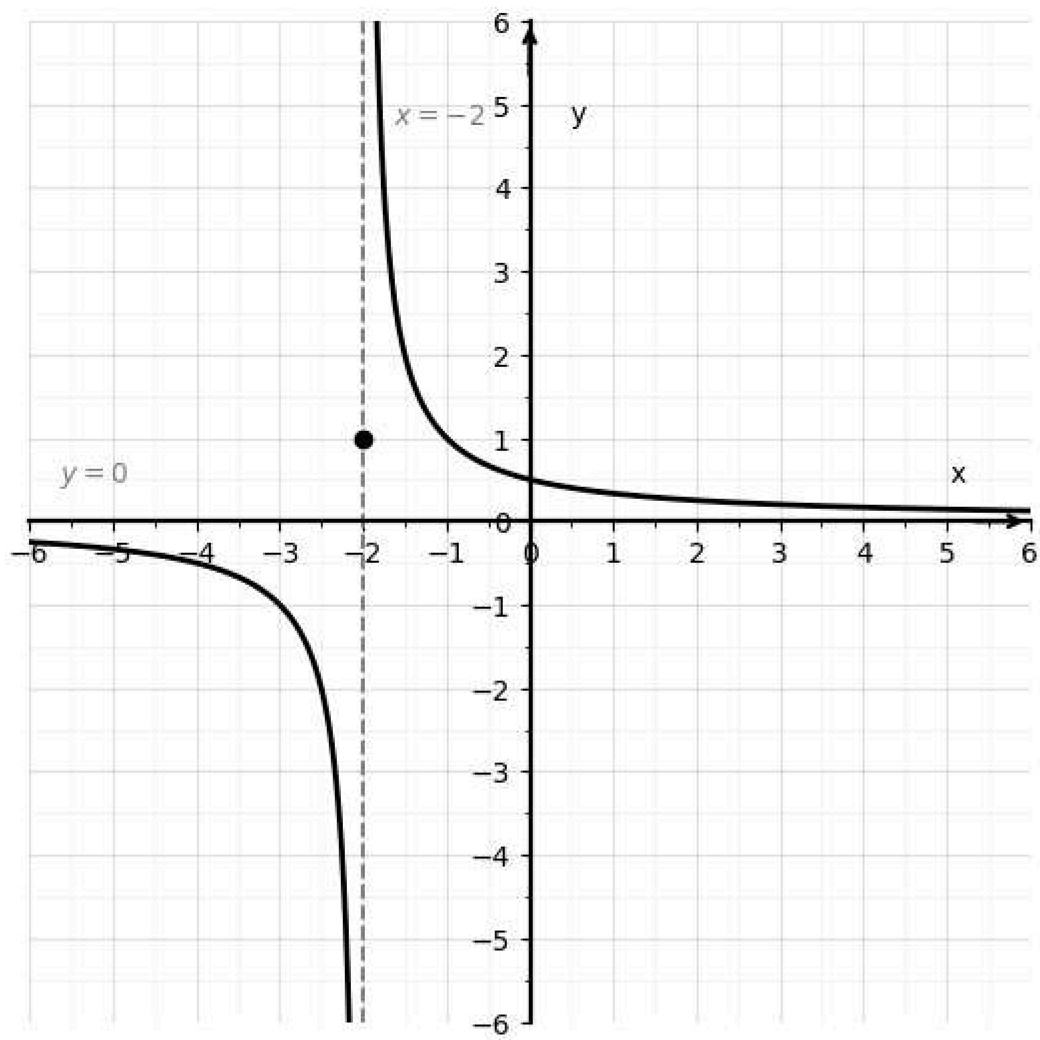
Since the one-sided limits are not equal, the two-sided limit does not exist

Because the limit does not exist, it cannot equal $f(-2) = 1$

Therefore the function is discontinuous at $x = -2$

$$\boxed{\text{Discontinuous at } x = -2}$$

$f(x) = \dfrac{1}{x+2}$ for $x \neq -2, \quad f(-2) = 1$



Question 24

The function is defined by

$$f(x) = \begin{cases} \dfrac{2x^2 - 5x - 3}{x - 3}, & x \neq 3, \\ 6, & x = 3, \end{cases}$$

with $a = 3$

Factor the numerator:

$$2x^2 - 5x - 3 = (2x + 1)(x - 3)$$

For $x \neq 3$

$$f(x) = 2x + 1$$

Two-sided limit

$$\lim_{x\to 3} f(x) = \lim_{x\to 3}(2x+1) = 7$$

But

$$f(3) = 6$$

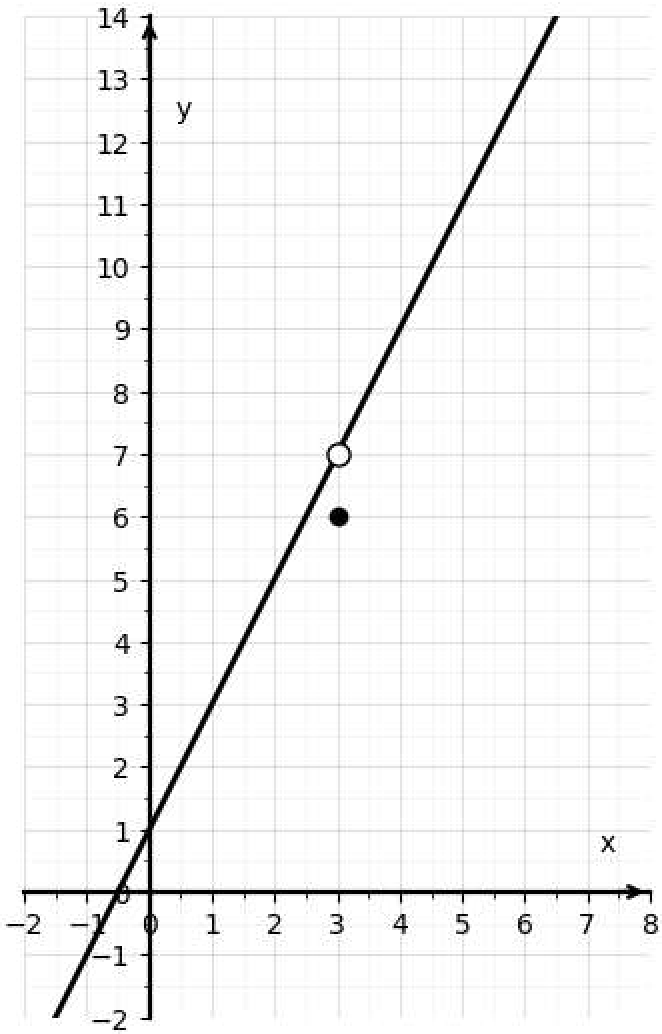
Since

$$\lim_{x\to 3} f(x) \neq f(3)$$

the function is discontinuous at $x = 3$

$$\boxed{\text{Removable discontinuity at } x = 3}$$

$$f(x) = \begin{cases} \dfrac{2x^2-5x-3}{x-3}, & x \neq 3 \\ 6, & x = 3 \end{cases}$$



Question 45

The function is defined by

$$f(x) = \begin{cases} x+2, & x<0, \\ e^x, & 0 \le x \le 1, \\ 2-x, & x>1 \end{cases}$$

Check continuity at the junction points

At $x = 0$

Left-hand limit

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} (x+2) = 2$$

Right-hand limit

$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} e^x = e^0 = 1$$

Since

$$\lim_{x\to 0^-} f(x) \ne \lim_{x\to 0^+} f(x)$$

the two-sided limit $\lim_{x\to 0} f(x)$ does not exist

Also

$$f(0) = e^0 = 1$$

$$\boxed{\text{Discontinuous at } x = 0}$$

At $x = 1$

Left-hand limit

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} e^x = e$$

Right-hand limit

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (2-x) = 1$$

Since

$$\lim_{x\to 1^-} f(x) \ne \lim_{x\to 1^+} f(x)$$

the two-sided limit $\lim_{x\to 1} f(x)$ does not exist

Also

$$f(1) = e^1 = e$$

$\boxed{\text{Discontinuous at } x = 1}$

$$f(x) = \begin{cases} x + 2, & x < 0, \\ e^x, & 0 \le x \le 1, \\ 2 - x, & x > 1 \end{cases}$$